The Law of Graphic Equalisation — Born Recovery

An Axiom-by-Axiom Note

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DRAFT — draft-2026-09-02.1 r2 (2 September 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) and to *The Law of Graphic Equalisation — Physics Grounding* (the physics companion), and inherits those papers' definitions, epistemic boundary, and level-of-claims; *The Law of Narrative Momentum* (the momentum paper) supplies the Narrative and the first law. It is an *attempt*, rated step by step, and not a peer-reviewed result. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This note checks, one axiom at a time, whether the change-substrate of Graphic Equalisation supplies the operational axioms from which the informational reconstructions of quantum theory derive the complex Hilbert space and the Born rule. The papers named above carry the surrounding framework, and are cited by number throughout rather than restated. The honest verb throughout is *recover*, not *derive*: the substrate reproduces the known rule within its own structures, given its admitted primitives and a contingent, self-located selection, rather than compelling it from necessity. Each step is rated solid, earned, plausible or contingent, and the residuals are named rather than hidden. Two findings carry the note. First, the substrate instantiates the operational-probabilistic core — convex states, affine effects, a transformation monoid with reversibles, composition, normalised probabilities — from the classical total-probability backbone and not from any quantum assumption. Second, two independent reconstructions have two *different* most-quantum axioms, and the substrate supplies each natively from a different one of its own principles: Hardy's continuity of reversible transformations from free continuation, and the purification of Chiribella, D'Ariano and Perinotti from entanglement as a superframe dimension. Complex over real is then not a gap to be closed by a forcing argument but the signature of a contingent selection, self-located exactly as the number of spatial dimensions is. The only claim made here is the *arrangement*; each component is individually known, and the uniqueness is supplied by the cited theorems.

1 The substrate, in one paragraph

Seven notions carry the mapping below. Each is defined in one of the three papers named above; this section fixes the references and adds nothing. A *frame* is a subgraph that runs (Graphic Equalisation's Definition 6.2), and frames compose by *nesting*, a graph being a subset of its container (its Proposition 6.6). A frame's state is its *Narrative*, the standing structure it carries forward (the momentum paper's Definition 2.2), and where a frame instantiates a direction — optional in general, present in this setting — that Narrative has a magnitude and a direction (the momentum paper's Definition 4.1); the state proper is the *internal* Narrative, as against

what another frame reads off it (its Definition 5.3). A *closure cycle* is the loop a realisation must close, the cycle being the unit of inferred time (Graphic Equalisation's Proposition 13.7), and *phase* is position on that cycle rather than a signed time (its Remark 13.8) — so an amplitude is $r\,e^{i\theta}$. A *realisation* is a collapse, a local write that picks one outcome from the admissible set and carries a weight (Graphic Equalisation's Definition 4.1), and realisations are strictly ordered, the order carried by the rules rather than by a clock (its Proposition 11.2). *Free continuation* is what a frame does when no rule fires, the information-preserving default (the momentum paper's Definition 5.2); *free* means information-preserving and not absent, since evaluating a rule is itself a write (Graphic Equalisation's Proposition 7.15), and all the mapping below uses is that free continuation is *injective*, hence reversible. Nothing is privileged: there is no god's-eye frame, and therefore no privileged phase origin and no privileged measurement basis (its Axiom -1). *Entanglement* is a superframe dimension: a subframe's local state is mixed because the purity is a dimension of the larger frame (the physics companion's Proposition 6.1 and its Corollary 6.2).

2 The claim, honestly sized

The substrate *recovers* the Born rule — reproduces it within its own structures, given its admitted primitives and a contingent, self-located selection — rather than *deriving* it as a forced necessity.

Remark 2.1 (Recovery is the honest verb, and the right one)*.* Derivation per se would compel Born from primitives. The framework forces almost nothing, and that is the point of a graph substrate: what survives every attempted removal is a container relation across which rules run (Graphic Equalisation's Remark 3.2), and essentially everything else is contingent rule-content, as with the number of spatial dimensions (the physics companion's Section 15) or the gauge group (its Remark 11.3). A contingent law can only be recovered given its rule-content, never derived as compelled. Some steps below are forced *given the premises* — the squared modulus, from no-privilege together with the additivity of decohered alternatives — while the premises themselves are admitted and contingent: complex, linear, self-similar.

Every distinctively-quantum ingredient is supplied by a substrate principle; the cited reconstruction theorems supply the uniqueness (Gleason, 1957; Hardy, 2001; Masanes and Müller, 2011; Chiribella et al., 2011). The only real claim is the *arrangement* — each component is individually known. That sizing is not modesty about the result but a statement of what kind of result it is, and it governs every rating below.

3 Level 0: the substrate is an operational-probabilistic theory

Before any reconstruction axiom applies, the substrate must be an operational-probabilistic theory (OPT): states in a convex set, effects, transformations, and a composition rule. That framework is a *precondition* for either axiom set, so the mapping has two levels, and this is the lower one.

The key that makes it tractable is seeing what the OPT convex structure *is*. The convexity of the state space and the affinity of effects are the *classical total-probability backbone* — total probability over ordered branchings — distinct from, and prior to, the quantum superposition linearity that Section 4 worries about. Every probabilistic theory has this backbone. The substrate has it by construction, because realisations carry weights, compose in order, and depend only on the present.

Proposition 3.1 (The substrate instantiates the operational-probabilistic core)**.** *Convex states, affine effects, a transformation monoid with reversibles, composition, and normalised probabilities are each supplied by the substrate's weighted, ordered realisations together with the history-optional, present-structure-acts first law (the momentum paper's Section 6; Graphic Equalisation's*

Corollary 12.5). The instantiation rests on the classical total-probability backbone and on no quantum assumption.

Ingredient by ingredient, with the substrate structure that supplies each. Four ratings are used here and in Section 4: *solid*, the substrate structure is the requirement's statement; *earned*, derived here from a substrate principle rather than read off one; *plausible*, a commitment of the right kind, with statement-match not shown; *contingent*, rule-content, admitted and self-located rather than supplied.

OPT requirement	**Supplied by**	**Rating**
Systems	Frame types: a subgraph together with an outcome set.	solid
States are well-defined equivalence classes	Statistics depend only on the current Narrative — history is optional and present structure acts, so there is no preparation-memory and operational equivalence holds.	solid
Convex state space	Mixing two preparations by a weighted coin-realisation: realise a λ-weighted coin, prepare P on one branch and Q on the other.	solid
Affine effects	Total probability over the coin–prepare–measure branching, $e(\lambda P + (1-\lambda)Q) = \lambda e(P) + (1-\lambda)e(Q)$.	solid
Transformations (monoid, with reversibles)	Rules compose, in order; free continuation is the identity, information-preserving and hence reversible.	solid
Composition	Subframe nesting; a joint preparation lives over the product outcome set. Existence is Level 0; the complex-specific tomography is the contingent layer of Section 5.	solid
Probabilities	Realisation weights in $[0, 1]$, normalised over a complete measurement.	solid

Remark 3.2 (The affinity is not the superposition linearity). The affinity of effects is the law of total probability applied to ordered realisations, and nothing about it is quantum. It is worth marking because the two linearities are easy to run together, and the argument would be circular if they were: Level 0 earns the classical backbone, and Section 4 must earn the superposition linearity separately, on other grounds.

What remains at this level is hygiene rather than core: a finite tomographic dimension, in the sense that a state is fixed by finitely many fiducial effects — plausible for finite frames by the aperture, unproven in general; topological closure and compactness of the convex state space; the full composition axioms, associativity of sequential and parallel composition and the interchange law, which should follow from the ordering and nesting of realisations but each need checking; and the no-restriction and effect-realisability niceties. None of these is the question whether the state space convex-linearises at all, which Level 0 answers; they are carried forward in Section 7.

4 The distinctively-quantum axioms, mapped

Two independent reconstructions are used. Hardy (2001) gives five axioms, of which classical probability theory satisfies the first four and fails the fifth. Chiribella et al. (2011) — Chiribella, D'Ariano and Perinotti, CDP below — give six principles that single out quantum theory over the real, the quaternionic and the classical. They have *different* most-quantum axioms, which is what makes the pair worth running rather than either alone.

Proposition 4.1 (Each reconstruction's crux axiom is supplied natively, from a different substrate principle)**.** *Hardy's quantum-selecting axiom is the continuity of reversible transformations*

between pure states; the substrate supplies it from free continuation, which is information-preserving and hence reversible, and whose phase is a continuous $U(1)$, so it moves continuously between pure states. CDP's characteristically quantum principle is purification; the substrate supplies it from entanglement as a larger-frame dimension, a mixed subframe state being the restriction of a pure superframe state. Neither is borrowed from the other, and neither was introduced for this purpose.

That two separate reconstructions' crux axioms are both already-present substrate structure, from two different principles, is the substantive positive finding of this note. The remaining correspondences are weaker and are marked as such:

Axiom	**Substrate structure**	**Rating**
Continuity of reversible transformations (Hardy; classical theory fails exactly this)	Free continuation: information-preserving, hence reversible, with a continuous cycle-phase.	solid
Purification (CDP; fails classically)	Entanglement as a superframe dimension: a mixed subframe state is the restriction of a pure superframe state.	solid
Phase- and context-independence	No privileged phase origin and no privileged frame: probability is invariant under global phase and the same across measurement bases.	earned
Probability	Weighted realisations; frequencies over an ensemble of collapses from one possibility structure.	plausible
Simplicity and ideal compression	Minimum description length as a native commitment (Graphic Equalisation's Remark 2.4) (Rissanen, 1978).	plausible
Subspaces	Subframe-as-frame: restriction to a subset of distinctions behaves as a smaller frame.	plausible
Perfect distinguishability	Distinctions are first-class; a state that is not completely mixed carries one.	plausible
Pure conditioning	Collapse of a pure Narrative yields a definite realised outcome.	plausible
Local tomography (Hardy's composite axiom; CDP's local distinguishability)	Subframe composition over the product outcome set — *contingent*, and the subject of Section 5.	contingent

The plausible ratings are genre-matches rather than statement-matches: the substrate has a commitment of the right kind, and that the specific axiom is the one the substrate enforces is not shown. Marking them so is the point of the table; a reader who upgrades one of them is disagreeing with a rating, which is the intended form of the disagreement.

The squared modulus is selected among the natural candidates by phase-averaged additivity, and made unique by Gleason, in the physics companion (its Proposition 7.1 and the argument following it); this note uses the result. What the framework supplies to Gleason (Gleason, 1957) are his premises — a vector-valued state, from the linearity step, and a context-independent measure, from the absence of a privileged rule set — and Gleason supplies what the framework does not.

Remark 4.2 (Two supporting identifications). The *inner product* is the cross-frame projection: one frame's Narrative read in another's basis, so it is not imported structure but the framework's own cross-frame reading. *Unitarity* is that a free continuation destroys no distinction — an information-preserving write rather than the absence of one, which is the right shape, since unitary evolution plainly does change the state and is reversible for exactly that reason. Both are identifications rather than derivations, and the rigorous step from "uncoupled" to full additivity and norm-preservation is the standing residual of Section 7.

5 Complex over real: contingent, not forced

Local tomography is the axiom that selects complex Hilbert space over real, and it is where a forcing argument would have to sit. Writing K for Hardy's fiducial degrees of freedom including normalisation, for two two-state systems:

Theory	K **single**	K **composite**	$K_A \cdot K_B$	**Local tomography**
Classical	2	4	4	holds (but fails continuity)
Real (rebit)	3	10	9	**fails** ($10 > 9$)
Complex (qubit)	4	16	16	**holds**

A closure-phase frame is *consistent with* complex amplitudes at one system — the Bloch sphere, $K = 4$, rather than the real disk's $K = 3$ — though a phase-less frame is equally constructible and the substrate requires neither. Read that way, a superframe, being a frame, is a complex closure-phase Narrative over the product outcome set, giving $K_{AB} = 16$: local-tomographic, complex.

Proposition 5.1 (Complex over real is a self-located selection, not a forced consequence)**.** *The framework forces almost nothing. Complex-over-real is contingent rule-content, exactly as the number of spatial dimensions is. Local tomography is therefore not a gap to be closed by a forcing argument; it is the signature of a contingent selection. A real-amplitude composition, $K_{AB} = 10$, is another admissible configuration. Our universe is the complex one, self-located (the physics companion's Proposition 14.2).*

Remark 5.2 (Why forcing is the wrong question)*.* Whether the substrate *forces* the complex composite is not the open question, and asking it would be an error of the framework's own kind rather than merely an unproven step. Seeking to force complex over real would contradict the framework's stance that it never forces contingent physics. The reading that survives is narrower and does real work: the admissible complex configuration yields $K_{AB} = 16$ and the admissible real one yields 10; a real composite would be a superframe that is not a closure-phase frame like its subframes, so real amplitudes cost self-similarity. That gives a direction and a reason without giving a compulsion, and it is the same status every other contingent constant in this programme has.

Given the contingent choice, the Born rule and the Hilbert structure follow via the mapping and the cited reconstructions — which is how the framework treats all physics: shapes follow from contingent rule-content, the rule-content itself never forced. Of the post-2011 reconstructions, Masanes and Müller (2011) obtain finite-dimensional quantum theory from five physical requirements on states and transformations, and theirs is the formulation this note would most like tested against, since those requirements are stated on single systems plus a composition rule — exactly the seam where the substrate's answer turns contingent rather than forced. Independent support for reading the selection as contingent comes from Höhn (2017) and Höhn and Wever (2017), whose rules on an observer's acquisition of information yield the correlation structure of qubits and rebits alike and so do not by themselves select complex over real. The rest of the reconstruction landscape is the physics companion's (its Section 17).

6 What this reconciles

Special and general relativity reconcile on this substrate as one change-budget spent two ways, with the invariance of c a definitional consequence rather than a postulate, and quantum collapse and relativity are joined at c as the in-frame event and the cross-frame map of one change-substrate (the physics companion's Proposition 4.1, its Corollary 4.2, and its Section 4.1). With Born *recovered* on the same primitives, Born, SR and GR stand reconciled on one substrate —

the reconciliation is that companion's, argued in its subsection on Born, SR and GR, and it is a reconciliation of the *conflict* at the level of shape, not a quantum theory of gravity: no field is quantised, no metric derived, no dimensionless constant fixed.

7 Residuals

What is open, unaddressed, and named:

1. **Level 0 hygiene.** Whether "present structure acts" genuinely delivers operational equivalence — states as equivalence classes of preparations with identical statistics — and whether the full OPT composition axioms (associativity of sequential and parallel composition, the interchange law) hold under the ordering and nesting of realisations. With them, a finite tomographic dimension, topological closure and compactness of the state space, and no-restriction and effect-realisability. These are hygiene rather than core, and plausible for finite frames; none is proven here.

2. **The linearity step.** The rigorous passage from "uncoupled free continuation" to full additivity and norm-preservation. The mechanism motivates linearity; it does not prove it, and the linearity of the imposed supergraph map is not handed over for free. This is the one posit the recovery rests on.

3. **The squaring argument.** Whether the phase-average selection is tight enough on its own, or whether the full Gleason content is needed to reach uniqueness. As the physics companion states it (its Proposition 7.1 and the argument following it) the phase average selects rather than proves, and Gleason is cited for the rest.

4. **Composition.** Under the natural reading the composite is local-tomographic and complex. Whether that reading is forced, permitted, or wrong is not settled here. The formulation of Masanes and Müller (2011) is the one this seam should be tested against, its requirements being stated on single systems plus a composition rule; that is exactly where the substrate's answer turns contingent rather than forced, and where the mapping is least confident that it is doing honest work.

5. **The plausible ratings.** Probability, simplicity and compression, subspaces, distinguishability and pure conditioning are genre-matches to native commitments, not statement-matches. Each would need tightening before the cited theorems could be applied without qualification.

8 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

1. [SHAPE] **The operational-probabilistic instantiation.** Fails if the substrate does not instantiate an OPT — if convex states, affine effects, a transformation monoid with reversibles, composition and normalised probabilities cannot all be had from weighted ordered realisations and the history-optional first law. Everything below is conditional on this, so it fails with it.

2. [DETAIL] **The Level-0 hygiene conditions.** Fails if the conditions of Section 7 — a finite tomographic dimension, topological closure and compactness of the state space, the full composition axioms — cannot be met for the frames this note is about. Each is a mechanism the note could lose and replace without losing the instantiation above.

3. [DETAIL] **Linearity from free continuation.** Fails if the step from uncoupled independent continuation to full additivity and norm-preservation cannot be made rigorous — in which case the vector structure is a posit rather than a recovery, and Gleason's first premise is imported rather than supplied.

4. [SHAPE] **The axiom mapping.** Fails if it does not go through for a named axiom: in particular if free continuation is not Hardy's continuous reversible transformation between pure states, or if entanglement as a superframe dimension is not CDP's purification. Failure at either of those two is worse than failure elsewhere, since they are the axioms that separate quantum from classical and are what Proposition 4.1 claims.

5. [DETAIL] **The squared modulus.** Fails if, given complex linearly superposing amplitudes and no privileged phase, the probability is not the squared modulus; and the phase-averaged selection argument of the physics companion (its Proposition 7.1 and the argument following it) fails separately if another natural candidate satisfies the phase-averaged additivity, leaving the whole weight on Gleason.

6. [DETAIL] **The composition seam.** Fails if the substrate's composition rule, tested against requirements stated on single systems plus a composition rule (Masanes and Müller, 2011), yields a composite that is neither the complex admissible configuration ($K_{AB} = 16$) nor the real one ($K_{AB} = 10$) — which would make the contingency verdict of Proposition 5.1 a verdict about the wrong set.

7. [SHAPE] **Complex over real is contingent.** Fails if the framework must *force* complex over real rather than admit both and self-locate — that is, if a substrate that admits a real-amplitude composition is thereby shown to be the wrong substrate, rather than one whose selection among admissible configurations is a matter of where the observer is.

Scope note (not a falsifier). This note is shape-level and not quantitative: it fixes no dimensionless constant, derives no field equation, and offers no theory of quantum gravity. Nor is it a derivation: the claim is the *arrangement*, and no step above is offered as compelling the Born rule from primitives.

References

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